

How It Works

This algorithm has two main parts: the `line()` function and the `rotatePoint()` function. The cube has eight points, one for each corner. The program stores these corners as `(x, y, z)` tuples in the `CUBE_CORNERS` list. These points also define the connections for the cube's edge lines. When all the points rotate in the same direction by the same amount, they give the illusion of a cube rotating.

```
1. """Rotating Cube, by Al Sweigart al@inventwithpython.com
2. A rotating cube animation. Press Ctrl-C to stop.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: large, artistic, math"""
5.
6. # This program MUST be run in a Terminal/Command Prompt window.
7.
8. import math, time, sys, os
9.
10. # Set up the constants:
11. PAUSE_AMOUNT = 0.1 # Pause length of one-tenth of a second.
12. WIDTH, HEIGHT = 80, 24
13. SCALEX = (WIDTH - 4) // 8
14. SCALEY = (HEIGHT - 4) // 8
15. # Text cells are twice as tall as they are wide, so set scaley:
16. SCALEY *= 2
17. TRANSLATEX = (WIDTH - 4) // 2
18. TRANSLATEY = (HEIGHT - 4) // 2
19.
20. # (!) Try changing this to '#' or '*' or some other character:
21. LINE_CHAR = chr(9608) # Character 9608 is '█'
22.
23. # (!) Try setting two of these values to zero to rotate the cube only
24. # along a single axis:
25. X_ROTATE_SPEED = 0.03
26. Y_ROTATE_SPEED = 0.08
27. Z_ROTATE_SPEED = 0.13
28.
29. # This program stores XYZ coordinates in lists, with the X coordinate
30. # at index 0, Y at 1, and Z at 2. These constants make our code more
31. # readable when accessing the coordinates in these lists.
32. X = 0
33. Y = 1
34. Z = 2
35.
36.
37. def line(x1, y1, x2, y2):
38.     """Returns a list of points in a line between the given points.
39.
40.     Uses the Bresenham line algorithm. More info at:
41.     https://en.wikipedia.org/wiki/Bresenham%27s_line_algorithm"""
42.     points = [] # Contains the points of the line.
43.     # "Steep" means the slope of the line is greater than 45 degrees or
44.     # less than -45 degrees:
45.
```